

Lesson 8

Topic: Introduction to DAX Basics & Calculated Columns vs. Measures

1. What does DAX stand for?

DAX = **Data Analysis Expressions**

It's the formula language used in Power BI, Power Pivot, and SSAS Tabular models.

2. DAX formula to sum the Sales column

Total Sales = SUM(Sales[Sales])

3. Difference between a Calculated Column and a Measure

- **Calculated Column:**
 - Computed row by row.
 - Stored in the model → increases file size.
 - Used like a regular column.
- **Measure:**
 - Calculated on the fly, based on context.
 - Not stored → more efficient.
 - Aggregates values dynamically.

4. Profit Margin using DIVIDE

Profit Margin = DIVIDE(SUM(Sales[Profit]), SUM(Sales[Sales]), 0)
(0 is returned if denominator = 0, avoiding errors).

5. COUNTROWS() in DAX

Counts the number of rows in a table or filtered table expression.

6. Measure: Total Profit

Total Profit = SUM(Sales[Sales]) - SUM(Sales[Cost])

7. Average Sales per Product

Avg Sales per Product = AVERAGEX(VALUES(Sales[Product]), SUM(Sales[Sales]))

👉 AVERAGEX iterates products and averages their sales.

8. IF() to tag products

High Profit Tag =

IF(SUM(Sales[Profit]) > 1000, "High Profit", "Low Profit")

9. Circular Dependency Error

Occurs when a calculated column or measure **depends on itself** (directly or indirectly), creating an infinite loop.

10. Row Context vs. Filter Context

- **Row Context:** Each row of a table has its own context (used in calculated columns, iterators like SUMX).
- **Filter Context:** Filters applied by slicers, visuals, or CALCULATE() that define the subset of data a measure evaluates.

11. YTD Sales using TOTALYTD()

YTD Sales = TOTALYTD(SUM(Sales[Sales]), Sales[Date])

12. Dynamic Measure (Sales vs. Profit vs. Margin)

Create a disconnected table MetricTable with values: *Sales, Profit, Margin*.

Selected Measure =

```
SWITCH(
    SELECTEDVALUE(MetricTable[Metric]),
    "Sales", SUM(Sales[Sales]),
    "Profit", SUM(Sales[Profit]),
    "Margin", DIVIDE(SUM(Sales[Profit]), SUM(Sales[Sales]))
)
```

13. Optimize with Variables

Instead of repeating SUM(), use VAR:

Profit Margin VAR =

VAR TotalSales = SUM(Sales[Sales])

VAR TotalProfit = SUM(Sales[Profit])

RETURN DIVIDE(TotalProfit, TotalSales, 0)

14. CALCULATE() to override filter

Example: ignore Product filter, show total Sales for all Products.

Sales All Products = CALCULATE(SUM(Sales[Sales]),
ALL(Sales[Product]))

15. Highest Sales Amount

Max Sales = MAX(Sales[Sales])

or, for total sales per product:

Max Product Sales = MAXX(VALUES(Sales[Product]), SUM(Sales[Sales]))